

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.4: TRIGONOMETRY 1

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your guide for constructing a complete, high-quality written answer to the driving question. Read every section carefully. Each section builds on the last — just like the way your understanding of trigonometry grew lesson by lesson, from that first photograph of the mast in Nakuru all the way to solving real height-and-distance problems across Kenya. Write your Final Explanation in your own words, using the prompts and exemplars below as your model. Your goal is to show that you truly understand how trigonometry works and why it matters — not just that you can follow steps.

Section 1 — Name the Problem and Introduce the Tool

Start by describing the anchoring phenomenon in your own words. What was the problem? Why couldn't we solve it with algebra or Pythagoras alone? What new mathematical tool did we need, and what is its name? Make sure you explain what trigonometry actually is — don't just say 'it is a branch of mathematics.' Tell the reader what it does.

At the start of our unit, we were shown a photograph of a Kenya Power transmission mast standing near a school in Nakuru County. A student was standing 40 m from the base and estimated the angle to the top of the mast was about 50° . The challenge was: can we find the height of the mast using only that one distance and that one angle — without climbing it, using a drone, or guessing?

This turned out to be impossible with the tools we already had. Pythagoras' theorem links the three sides of a right-angled triangle to each other, but it needs at least two sides to work. Here we only had one side and one angle. Algebra alone could not connect an angle to a length. We had a real knowledge gap — and that gap is exactly what trigonometry fills.

Trigonometry is the branch of mathematics that studies the relationships between the angles and the side lengths of right-angled triangles. It gives us three ratios — sine, cosine, and tangent — that act as a reliable bridge between any angle in a right-angled triangle and the ratio of two of its sides. Once you know one angle and one side, those ratios let you calculate any other side. That is the tool that lets us find the height of the Nakuru mast without ever leaving the ground.

Section 2 — Explain the Three Trigonometric Ratios

Explain what sine, cosine, and tangent are. Use a clearly

In any right-angled triangle, once you choose one of the acute angles (call it θ), the three sides have specific names:

- The hypotenuse is always the longest side — the one directly opposite the right angle.

labelled right-angled triangle in your explanation — you can draw one or describe one using words. Define the opposite side, adjacent side, and hypotenuse relative to a chosen angle θ. Write out the three ratios as fractions and include the SOH-CAH-TOA memory aid. Then show, step by step, how you would use the tangent ratio to calculate the height of the Nakuru mast (40 m from the base, angle of elevation = 50°). Show your working clearly.	– The opposite side is the side directly across from angle θ. – The adjacent side is the side next to angle θ (that is not the hypotenuse). The three trigonometric ratios are: – $\sin \theta = \text{opposite} \div \text{hypotenuse}$ – $\cos \theta = \text{adjacent} \div \text{hypotenuse}$ – $\tan \theta = \text{opposite} \div \text{adjacent}$ A useful memory aid is SOH-CAH-TOA: Sin = Opposite/Hypotenuse Cos = Adjacent/Hypotenuse Tan = Opposite/Adjacent These ratios are fixed for any given angle — they do not change no matter how big or small the triangle is. That is what makes them so powerful. Now let's solve the Nakuru mast problem: – The student stands 40 m from the base of the mast — this is the adjacent side. – The angle of elevation to the top is 50° — this is angle θ. – The height of the mast is unknown — this is the opposite side. Because we have the adjacent side and want the opposite side, we use tangent: $\tan 50^\circ = \text{opposite} \div \text{adjacent}$ $\tan 50^\circ = h \div 40$ $h = 40 \times \tan 50^\circ$ $h = 40 \times 1.1918$ $h \approx 47.67 \text{ m}$ The mast is approximately 47.7 m tall — calculated entirely from the ground, using one distance and one angle.
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Section 3 — Angles of Elevation and Depression, and Special Angles	
Explain what an angle of elevation is and what an angle of depression is. How are they related to each other when an observer and an object are at different heights? Use a real Kenyan example for each (for example: looking up at the top of a building in Nairobi's CBD for elevation; looking down at a dhow from the top of Fort Jesus	An angle of elevation is the angle you measure upward from the horizontal when you look up at an object above you. For example, if you stand on Tom Mboya Street in Nairobi and look up at the top of a tall building under construction, the angle between the horizontal ground line and your line of sight to the top of the building is the angle of elevation. An angle of depression is the angle you measure downward from the horizontal when you look down at an object below you. For example, a surveyor standing at the top of Fort Jesus in Mombasa looking down at a dhow anchored in the harbour measures the angle of depression from the horizontal to the line of sight to the dhow. Importantly, the angle of elevation from point A to point B is always equal to the angle of depression from point B to point A. This is because the two horizontal lines are parallel, and the line of sight acts as a transversal — forming alternate interior angles that are

in Mombasa for depression). Then explain what special angles are (30°, 45°, 60°) and why their exact trigonometric values are useful — include at least two of these exact values in a small table.

equal.

Special angles — 30°, 45°, and 60° — are angles whose trigonometric values can be written as exact fractions or surds (no rounding needed). This makes calculations faster and perfectly accurate.

Angle	sin	cos	tan
30°	1/2	$\sqrt{3}/2$	1/√3
45°	1/√2	1/√2	1
60°	$\sqrt{3}/2$	1/2	√3

For instance, if a roof beam in a house in Kisumu makes a 45° angle with the ceiling and the horizontal span is 3 m, you can find the beam length exactly: beam = $3 \div \cos 45^\circ = 3 \div (1/\sqrt{2}) = 3\sqrt{2}$ m — no calculator rounding required.

Section 4 — Connecting Trigonometry to Real Kenyan Problems

This is your 'so what?' section. Give at least THREE distinct real-life examples — set in Kenya — where trigonometry is used to find an unknown height or distance. For each example, name the context, identify which trigonometric ratio applies, and write the equation you would set up (you do not need to solve all of them fully, but you must set up the correct equation). Think across different fields: construction, agriculture, surveying, navigation, or everyday life.

Trigonometry is not just a classroom exercise — it is used every day by Kenyans working in construction, agriculture, surveying, and beyond. Here are three real examples:

1. CONSTRUCTION — Roof design in Eldoret

A builder in Eldoret is designing a house with a roof that must rise 2.4 m above the ceiling joists over a horizontal span of 4 m. To cut the rafters at the correct angle, the builder uses:

$$\tan \theta = \text{opposite} \div \text{adjacent} = 2.4 \div 4 = 0.6$$

$$\theta = \tan^{-1}(0.6) \approx 31^\circ$$

Knowing this angle, the builder can cut every rafter accurately without measuring each one individually.

2. SURVEYING — Measuring the width of River Tana

A land surveyor working near Garissa cannot cross the flooded River Tana. Standing at point A on one bank, she sights a tree at point B directly across the river. She walks 80 m along the bank to point C and measures the angle ACB as 62°. Using the tangent ratio:

$$\tan 62^\circ = \text{AB} \div \text{AC} \rightarrow \text{AB} = 80 \times \tan 62^\circ \approx 80 \times 1.8807 \approx 150.5 \text{ m}$$

The river is approximately 150.5 m wide — measured entirely from one bank.

3. AGRICULTURE — Estimating the height of a Kericho tea factory chimney

A health inspector in Kericho needs to estimate the height of a tea factory chimney to check whether smoke is dispersing safely. Standing 60 m from the base on flat ground, she measures the angle of elevation to the top as 38°:

$$\tan 38^\circ = h \div 60 \rightarrow h = 60 \times \tan 38^\circ \approx 60 \times 0.7813 \approx 46.9 \text{ m}$$

Without any climbing equipment, she calculates the chimney is approximately 46.9 m tall and can proceed with her safety assessment.

Section 5 — Answer the Driving Question Directly

Now write a clear, confident, well-structured answer to the driving question: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance — and how does this help Kenyans solve real problems in construction, surveying, and everyday life?' Your answer should summarise everything you have learned across the 8 lessons. Use specific mathematical vocabulary (sine, cosine, tangent, angle of elevation, angle of depression, SOH-CAH-TOA, hypotenuse, opposite, adjacent). Aim for at least one full paragraph of explanation and one paragraph connecting to real life. This is your moment to show full understanding.

We can find any unknown height or distance in a right-angled triangle — given one side and one acute angle — by using the three trigonometric ratios: sine, cosine, and tangent. These ratios (remembered using SOH-CAH-TOA) express a fixed relationship between an angle and the ratio of two sides of the triangle. Tangent is especially useful for height-and-distance problems because it connects the opposite side (the unknown height) to the adjacent side (the known horizontal distance) through the angle of elevation or depression. Once you identify which two sides are involved and which angle is known, you set up the correct ratio, substitute the known values, and solve algebraically for the unknown side. Special angles such as 30° , 45° , and 60° have exact trigonometric values, so calculations involving them require no rounding. For all other angles, a scientific calculator gives the decimal approximation needed.

This matters enormously for Kenyans in many fields of work and life. Construction workers in Eldoret and Nairobi use trigonometric ratios to calculate roof angles, rafter lengths, and safe inclines for ramps — ensuring buildings are structurally sound without expensive guesswork. Surveyors measuring land boundaries or the width of flooding rivers like the Tana use tangent to calculate distances that cannot be physically walked. Health and safety inspectors use angles of elevation to estimate chimney heights at factories in Kericho or Thika. Even a mobile phone mast engineer planning a new tower in Nakuru uses trigonometry to determine how tall the mast must be to cover a given area. In every case, the core idea is the same: a right-angled triangle, one known side, one known angle, and a trigonometric ratio are all you need to find what was previously unreachable. Trigonometry turns the inaccessible into the calculable — and that is why it is one of the most practically powerful tools in all of mathematics.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of Trigonometric Ratios (sine, cosine, tangent)	Accurately defines all three ratios using opposite, adjacent, and hypotenuse. Correctly applies SOH-CAH-TOA and selects the appropriate ratio for each problem. Clearly explains why each ratio is chosen with reference to the known and unknown sides.	Defines all three ratios correctly and applies the right ratio in most situations. Minor errors in side identification or ratio selection in complex contexts, but the core understanding is clear.	Defines one or two ratios correctly but confuses opposite and adjacent, or selects the wrong ratio for the problem. SOH-CAH-TOA is mentioned but not applied consistently.
Solving Height-and-Distance Problems (including the Nakuru mast)	Sets up the trigonometric equation correctly, substitutes values accurately, and shows all algebraic working steps clearly. Arrives at a correct, rounded answer with appropriate units. Solves the Nakuru mast problem fully and correctly ($h \approx 47.7$ m).	Sets up the equation correctly and shows most working steps. May make a minor arithmetic or rounding error but the method is sound. The Nakuru mast problem is attempted with the correct approach.	Attempts to set up an equation but makes errors in identifying the correct ratio or substituting values. Working is incomplete or difficult to follow. The Nakuru mast problem is set up incorrectly or not attempted.

Angles of Elevation, Angles of Depression, and Special Angles	Clearly and accurately defines angles of elevation and depression, correctly explains their relationship (alternate interior angles / equal values), and uses at least one Kenyan real-life context for each. Provides a correct table of exact values for at least two special angles and explains why exact values are useful.	Correctly defines angles of elevation and depression and notes they are equal in corresponding situations. Provides at least one real-life Kenyan example. Gives some correct exact values for special angles but the explanation of their usefulness may be brief.	Defines angles of elevation or depression but not both accurately, or confuses which is which. May not explain the relationship between them. Special angles are mentioned but exact values are incomplete or contain errors.
Application to Real Kenyan Contexts	Provides at least three distinct, well-described real-life Kenyan examples across different fields (e.g., construction, surveying, agriculture). Each example clearly identifies the context, the correct trigonometric ratio, and a correctly set-up equation. Examples are specific to Kenyan places, industries, or people.	Provides two or three Kenyan real-life examples. Most examples include a correct equation set-up and a relevant context. One example may be generic or lack a fully set-up equation.	Provides one or two examples, which may be vague or not clearly set in a Kenyan context. Equations are missing or set up incorrectly. The connection between the mathematics and the real-world application is unclear.
Direct Answer to the Driving Question	The final section directly, confidently, and completely answers the driving question using precise mathematical vocabulary (sine, cosine, tangent, angle of elevation, angle of depression, hypotenuse, opposite, adjacent, SOH-CAH-TOA). Connects mathematical concepts to at least two Kenyan real-world fields. The response demonstrates genuine understanding, not just memorised steps.	The final section answers the driving question adequately and uses most of the required vocabulary. The connection to Kenyan real-life contexts is made but may be brief or cover only one field. The response shows solid understanding with minor gaps.	The final section attempts to answer the driving question but is vague, incomplete, or relies heavily on restating definitions without demonstrating understanding. Few or no key vocabulary terms are used correctly. The connection to real life is missing or superficial.